

ADOMAS VALANTINAS, PhD

Senior Fellow, Leverhulme Centre for Life in the Universe, University of Cambridge, UK

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RESEARCH INTERESTS

I study the geological and climatic evolution of Mars, with a focus on aqueous alteration, iron oxide mineralogy, and the habitability of ancient crustal environments. My research combines orbital hyperspectral and multispectral remote sensing from ESA and NASA missions, laboratory spectroscopy and characterization of alteration minerals, machine learning approaches to large planetary datasets, and terrestrial analog fieldwork in volcanic and hydrothermal settings.

RESEARCH APPOINTMENTS

Senior Fellow 09/26–present

Leverhulme Centre for Life in the Universe (LCLU), University of Cambridge, UK

Host: Prof. Nicholas Tosca

Focus: experimental aqueous geochemistry of iron minerals and potential biosignatures on Mars

SNSF Postdoc.mobility Fellow 09/23–08/26

Brown University, USA (two years) and ETH Zurich, Switzerland (one year)

Advisors: Prof. Jack Mustard (Brown) and Prof. Cara Magnabosco (ETH)

Laboratory skills: X-ray crystallography, geochemistry, VNIR/mid-IR spectroscopy of aqueous alteration minerals

Computational skills: machine learning for hyperspectral and multispectral remote sensing

Postdoctoral Researcher 12/22–03/23

University of Bern, Switzerland

Advisor: Dr. Antoine Pommerol

Laboratory skills: VNIR spectroscopy of iron oxides

Graduate Research Assistant 10/18–11/22

University of Bern, Switzerland

Remote sensing: geological and spectral analysis of multispectral camera data

Laboratory: bi-directional reflectance measurements of Martian soil simulants

Numerical modeling: reflectance modeling using Hapke theory

Master's Degree Researcher 10/15–10/17

University of Copenhagen, Denmark

Laboratory: Mastcam-Z calibration target development for NASA's Perseverance rover

EDUCATION

Ph.D. Astronomy and Physics 10/18–11/22

University of Bern, Switzerland

Thesis: *Bright and Dusty Regions of Mars: New Insights from Experiments and Orbital Color Imaging*

Advisor: Prof. Nicolas Thomas

M.Sc. Geophysics 10/15–10/17

University of Copenhagen, Denmark

Advisor: Prof. Kjartan Kinch

B.Sc. Astronomy and Physics 09/11–07/15

Vilnius University, Lithuania

Advisor: Dr. Audrius Bridzius

GRANTS AND AWARDS

LCLU Senior Fellowship, Leverhulme Centre for Life in the Universe, University of Cambridge (PI; \$374,000)	09/26–08/30
SNSF <i>Postdoc.mobility</i> fellowship, two years at Brown University and one year at ETH Zurich (PI; \$301,000)	09/23–08/26
American Philosophical Society Baruch Blumberg Grant for Astrobiology, “Decoding Mars Through Iceland: Ferrihydrite as a Key to Past Habitability and Potential Biosignatures”, for field work and conference attendance in Iceland (PI; \$8,500)	07/25
CIFAR seed funding, “Interfaces of Possibility: Modeling Boundary Ecosystems Across Planetary Systems” (Co-PI; \$7,200)	05/25
Lunar and Planetary Institute (LPI) Career Development Award (meeting canceled due to COVID)	03/20
American Geophysical Union (AGU) student travel award (\$2,000)	12/17

MISSION EXPERIENCE

ESA Trace Gas Orbiter (TGO) / CaSSIS , Science Team collaborator and operations planner (PI: N. Thomas). Planned ~10 months of CaSSIS observations (11% of total mission time). Science Theme lead for Hydrothermal Processes and for Geologic Contacts & Stratigraphy. Contributed 31,000 targets to the CaSSIS targeting database (60% of all team targets).	10/18–present
NASA Mars Reconnaissance Orbiter (MRO) / HiRISE , Science Team collaborator (PI: A. McEwen). Regular contributor to the HiRISE targeting database.	06/20–present
NASA Double Asteroid Redirection Test (DART) , guest participant in the DART Borders program (PI: A. Cheng). Observed team meetings and took part in discussions.	09/22–01/23
NASA Mars 2020 / Mastcam-Z , Science Team collaborator (PI: J. Bell).	10/15–10/17

FIRST-AUTHOR PUBLICATIONS

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1. **Valantinas, A.**, Mustard, J., Tullo, A. and Head, J.W. (in review at *EPSL*). An exposed magmatic feeder system for the regional olivine unit in Nili Fossae, Mars. *ESS Open Archive*. doi:10.22541/essoar.176235473.32635438/v1
 2. **Valantinas, A.**, Rajsic, A., Mustard, J., Tullo, A. et al. (in review at *GRL*). Partially altered olivine in the western rim of Jezero crater, Mars: preserved pre-impact Noachian basement. *ESS Open Archive*. doi:10.22541/essoar.175579625.58968122/v2
 3. **Valantinas, A.**, Mustard, J., Chevrier, V. et al. (2025). Detection of ferrihydrite in Martian red dust records ancient cold and wet conditions on Mars. *Nature Communications*, 16, 1712. doi:10.1038/s41467-025-56970-z
 4. **Valantinas, A.**, Thomas, N., Pommerol, A. et al. (2024). Evidence for transient morning water frost deposits on the Tharsis volcanoes of Mars. *Nature Geoscience*, 17, 608–616 (cover article). doi:10.1038/s41561-024-01457-7
 5. **Valantinas, A.**, Becerra, P., Pommerol, A. et al. (2021). CaSSIS color and multi-angular observations of Martian slope streaks. *Planetary and Space Science*, 209, 105373. doi:10.1016/j.pss.2021.105373
 6. **Valantinas, A.** and Schultz, P.H. (2020). The origin of neotectonics on the lunar nearside. *Geology*, 48, 649–653. doi:10.1130/G47202.1
 7. **Valantinas, A.**, Kinch, K.M. and Bridzius, A. (2018). Low crater frequencies and low model ages in

lunar maria: Recent endogenic activity or degradation effects? *Meteoritics & Planetary Science*, 53, 826–838. doi:10.1111/maps.13033

CO-AUTHOR PUBLICATIONS

1. Munaretto, G., (...) **Valantinas, A.** (2025). Phase reddening of Phobos and Deimos from TGO/CaSSIS observations. *Astronomy & Astrophysics*, 700, L1. doi:10.1051/0004-6361/202555720
2. Bickel, V. and **Valantinas, A.** (2025). Streaks on Martian slopes are dry. *Nature Communications*, 16, 4315. doi:10.1038/s41467-025-59395-w
3. Thomas, N., (...) **Valantinas, A.** et al. (2025). Seasonal and short timescale changes on the Martian surface: Multi-spacecraft perspectives. *Space Science Reviews*, 221, 3. doi:10.1007/s11214-024-01128-4
4. Mills, M.M., (...) **Valantinas, A.** (2024). A global dataset of pitted cones on Mars. *Icarus*, 418, 116145. doi:10.1016/j.icarus.2024.116145
5. Pommerol, A., (...) **Valantinas, A.** et al. (2022). In-flight radiometric calibration of the ExoMars TGO Colour and Stereo Surface Imaging System. *Planetary and Space Science*, 105580. doi:10.1016/j.pss.2022.105580
6. Munaretto, G., (...) **Valantinas, A.** et al. (2022). Multiband photometry of Martian Recurring Slope Lineae (RSL) and dust-removed features at Horowitz crater, Mars from TGO/CaSSIS color observations. *Planetary and Space Science*, 105443. doi:10.1016/j.pss.2022.105443
7. Dundas, C.M., (...) **Valantinas, A.** et al. (2021). Active Mars: A dynamic world. *JGR: Planets*, 126, e2021JE006876. doi:10.1029/2021JE006876

STUDENT SUPERVISION

Luis Grimaldo Estrada , M.Sc. Data Science, ETH Zurich. Course project and upcoming M.Sc. thesis on machine learning for Mars orbital data. Main supervisor.	09/26–present
Ari Antje Hölscher , M.Sc. summer research project, ETH Zurich. Machine learning on Mars image data.	Summer 2026
Chen Orenshtein , B.Sc. summer research project, ETH Zurich. Machine learning on Mars image data.	Summer 2026
Andrei Girjoaba , M.Sc.-level research assistant (Data Science), ETH Zurich. Machine learning for Mars remote sensing data. Co-supervised with Prof. Cara Magnabosco.	03/26–09/26
Guilliat Corboz , B.Sc. thesis, ETH Zurich. “Searching for Hydrothermal Systems on Mars: Using Polygonal Ridges to Map Subsurface Habitability”. Main supervisor; co-supervised with Prof. Cara Magnabosco.	02/26–06/26
Rushil Kukreja , high school research mentee, Brown University. Spectral analysis of Martian orbital data.	09/23–05/25

TEACHING

Laboratory teaching assistant, University of Bern. Instructed students in optics and thermodynamics laboratory courses.	10/18–11/22
Private tutor in physics and mathematics for high school students, Lithuania.	11/17–09/18

PROFESSIONAL SERVICE AND LEADERSHIP

Member, American Geophysical Union. Proposed and co-chaired, with Dr. Mike Wong, the interdisciplinary session “The Emergence of Life as a Planetary Process” at the 2025 AGU Fall Meeting, bridging the planetary science, astrobiology and geochemistry communities. Chairing a session again at the 2026 Fall Meeting.	12/17–present
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Led the Astrobiology and Origin of Life (OLA) reading group at Brown University: biweekly seminars bringing together undergraduates, graduate students, postdocs and faculty from Physics and from Earth, Environmental & Planetary Sciences to discuss seminal books and papers in astrobiology.	01/24–05/25
Proposal reviewer, NASA Mars Data Analysis Program (MDAP).	01/24
Dwornik Award judge for student presentations, Lunar and Planetary Science Conference.	03/23 & 03/25
Peer reviewer for <i>Earth-Science Reviews</i> , <i>Geology</i> , <i>Geophysical Research Letters</i> , <i>Journal of Geophysical Research</i> , <i>Planetary Science Journal</i> and <i>Icarus</i> .	09/21–present
Member, Geological Society of America.	09/21–present

OUTREACH AND PUBLIC ENGAGEMENT

Expert interviews with Associated Press, CNN, The Guardian, Royal Society of Chemistry (UK), and Swiss and Lithuanian national TV about the ferrihydrite finding on Mars. Guest on the Planetary Society's <i>Planetary Radio</i> podcast, "Why is Mars red? A new clue to the history of habitability in Martian dust". ~500,000 YouTube views across multiple channels.	02/25
Expert interviews with CNN, BBC Global (live TV), ABC Australia (live radio), USA Today, New Scientist and others about water frost on Martian volcanoes. ~80,000 Reddit upvotes (front page of reddit.com). Research image selected for the cover of <i>Nature Geoscience</i> , July 2024.	06/24
Radio DJ and science communicator at NTS.live and RadioVilnius.live, curating shows that combine experimental electronic music with planetary science for non-academic audiences.	06/23–present
Artistic collaboration with Matteo Taramelli for the exhibition "Potentially Uninhabitable Worlds", Dreiviertel Gallery, Bern, Switzerland.	06/21
Interview with <i>Ciel et Espace</i> (French popular science magazine) about active lunar tectonics.	06/20

SELECTED TALKS

"Why is Mars red? Recent advances in solving the century-old problem in planetary science", MIT/Harvard Planetary meeting, MIT, USA.	09/24
"Are Martian volcanoes dormant?", Planetary Lunch Bunch, Brown University, USA.	05/24
"Neotectonics on the Lunar Nearside", Observatoire de la Côte d'Azur, Nice, France.	10/17

WORKSHOPS AND SUMMER SCHOOLS

NASA Planetary Science Summer School (PSSS) for mission design, JPL, Pasadena, USA. Helped design a concept mission to Neptune's moon Triton.	08/25
CIFAR and SNSF "Forum on Radical Interdisciplinarity", Geneva, Switzerland.	03/25
CIFAR "Earth 4D: Subsurface Science & Exploration" program meeting, Quebec City, Canada.	06/24
European Astrobiology Institute, "Life in the Sub-Surface: Habitats, Species, Metabolism and Survival Strategies", Azores, Portugal.	10/23
European Space Agency workshop "Mars V", Les Houches, France.	04/22
GeoPlaNet analogue field winter school, "Fluid-Rock Interactions in the Solar System", Nantes, France.	11/18

Nordic Astrobiology Network field summer school, “Impacts and their Role in the Evolution of Life”, Saaremaa, Estonia.	07/17
Nordic Astrobiology Network field summer school, “Biosignatures and the Search for Life on Mars”, Iceland.	06/16

FIELD EXPERIENCE

Iceland: field project funded by the Baruch Blumberg Grant for Astrobiology.	07/25
Block Island, USA: characterization of iron-rich springs (Brown University field trip).	03/24
Azores, Portugal (Pico, Faial, São Miguel and Terceira): sampling of volcanic and hydrothermal environments, funded by SNSF. Ferrihydrite samples from Terceira were published in Valantinas et al. (2025).	11/23
Iceland: adenosine triphosphate (ATP) analysis and collection of volcanic tephra, Nordic Astrobiology Network summer school.	06/16